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# **KumoRFM: A Foundation Model for In-Context Learning on Relational Data**

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# Traditional Machine Learning Tasks

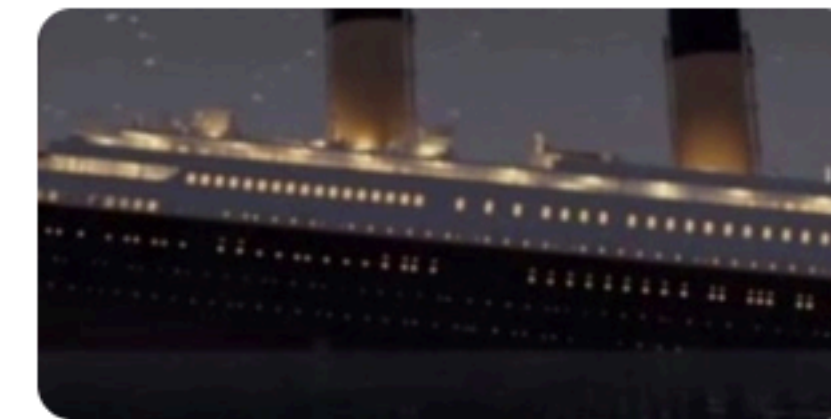
KAGGLE · GETTING STARTED PREDICTION COMPETITION · ONGOING

Submit Prediction

...

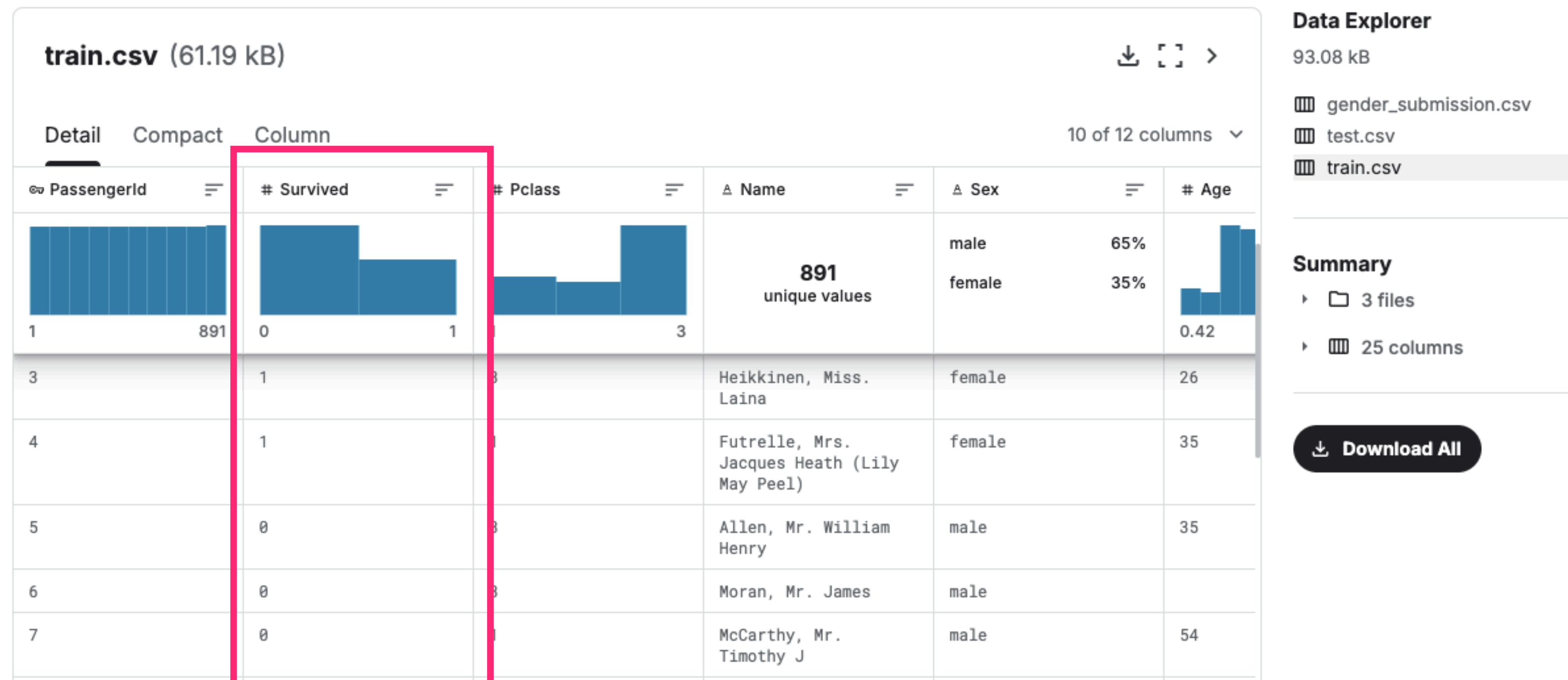
## Titanic - Machine Learning from Disaster

Start here! Predict survival on the Titanic and get familiar with ML basics



### Dataset Characteristics

- Structured Data Schema
- Clear Supervision



### Typical workflow

- EDA
- Data cleaning
- Data engineering
- Fit model on training data
- Hyperparameter tuning
- Deploy!

# Why not LLMs?

## LLMs on Textified Structured Data

(Hegselmann et al., 2023; Fang et al., 2024; Zhao et al., 2023; Wydmuch et al., 2024; Zhang et al., 2024)

- Transform the structured data into textual format (JSON, natural language, etc)
- Feed them as input prompt with a question to an LLM
- LLM reasons over the prompt using its pre-trained language understanding



- Brittle prompt engineering
- Suffer from long context windows
- Hallucinations
- Struggle to reason over numerical values
- ...

**Traditional ML models:** statistical estimation problem

**LLMs:** language reasoning problem (via next token prediction)

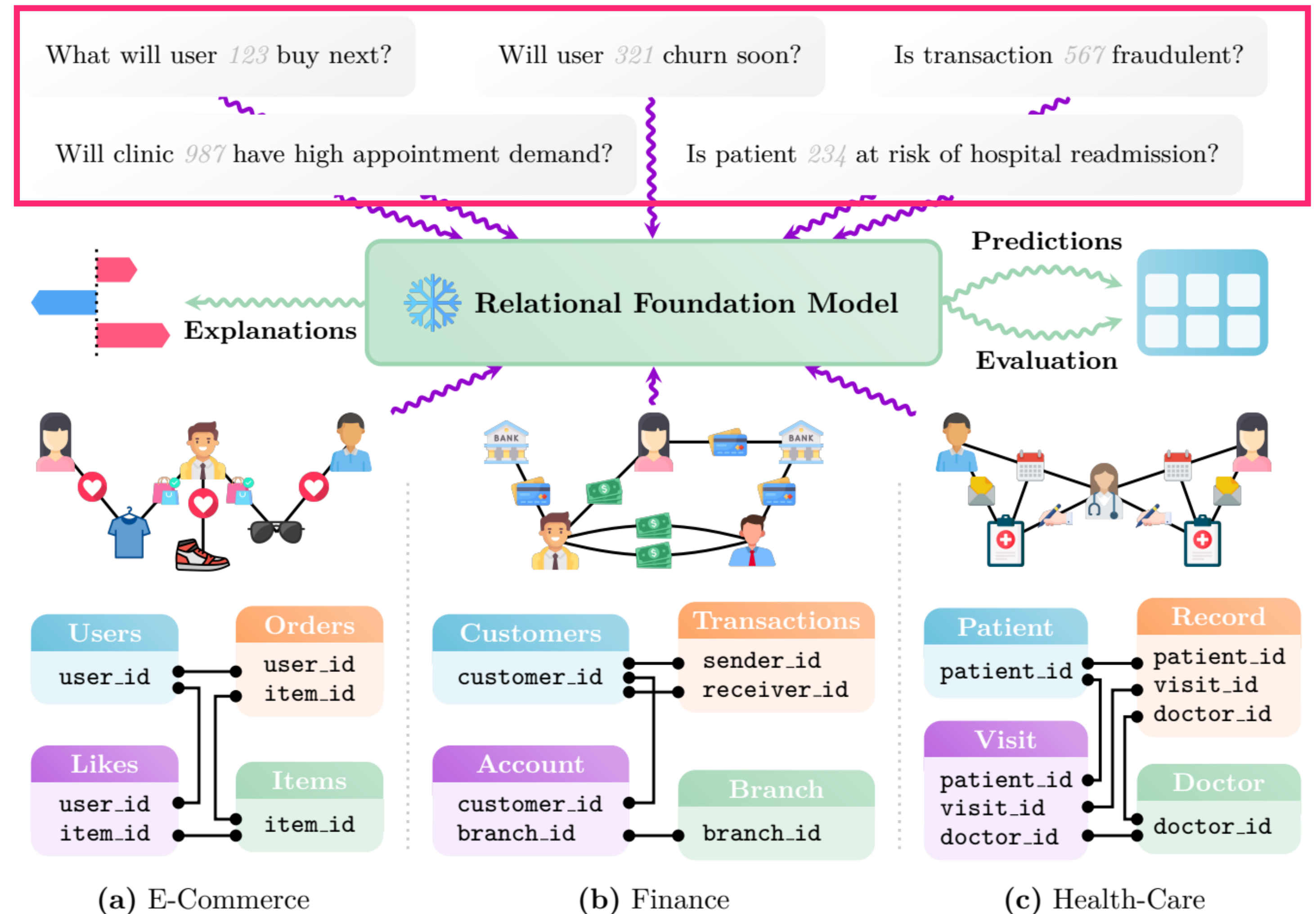
# RFMs are filling the hole left by LLMs

LLMs replaces difficult,  
expensive NLP tasks



RFMs replaces difficult,  
expensive predictive modeling  
task

RFM is not competing with LLMs,  
it's a complementary to LLMs



# Comparison with LLMs

## Large Language Model (LLM):

Doubt thou the stars are fire,  
Doubt that the sun doth move,  
Doubt truth to be a liar,  
But never doubt I love...

Text

**Natural Language Prompt**  
Context: Unstructured Docs

Common-sense  
in-context  
reasoning

Text output

## KumoRFM:



**Relational Prompt**  
Context: Relational Data

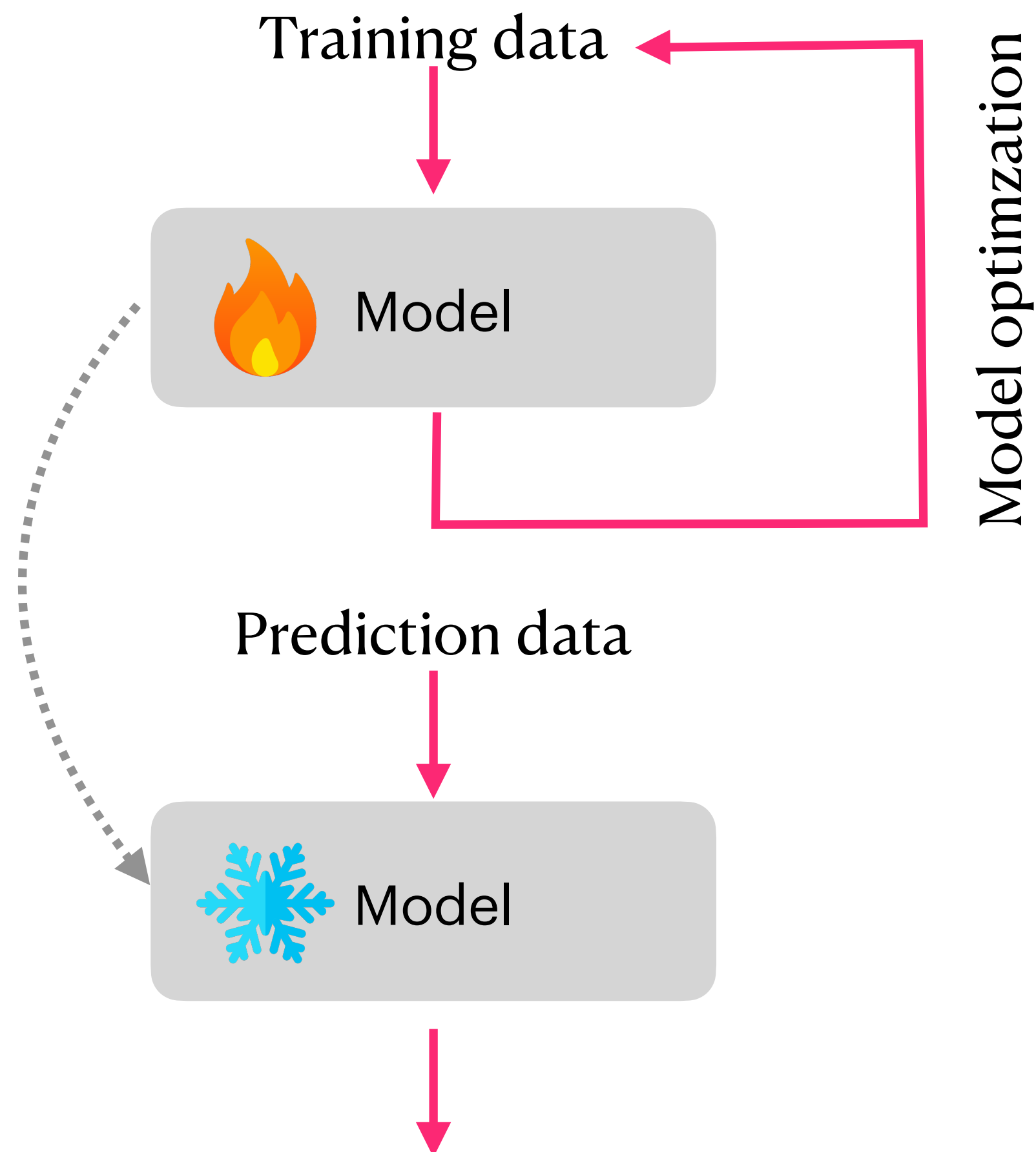
**Predict churn for active customers:**  
*PREDICT* COUNT(Sales.\*, 0, 30, days)=0  
*FOR EACH* Customers.ID  
*WHERE* COUNT(Sales.\*, -90, 0, days)>0

Relational  
reasoning

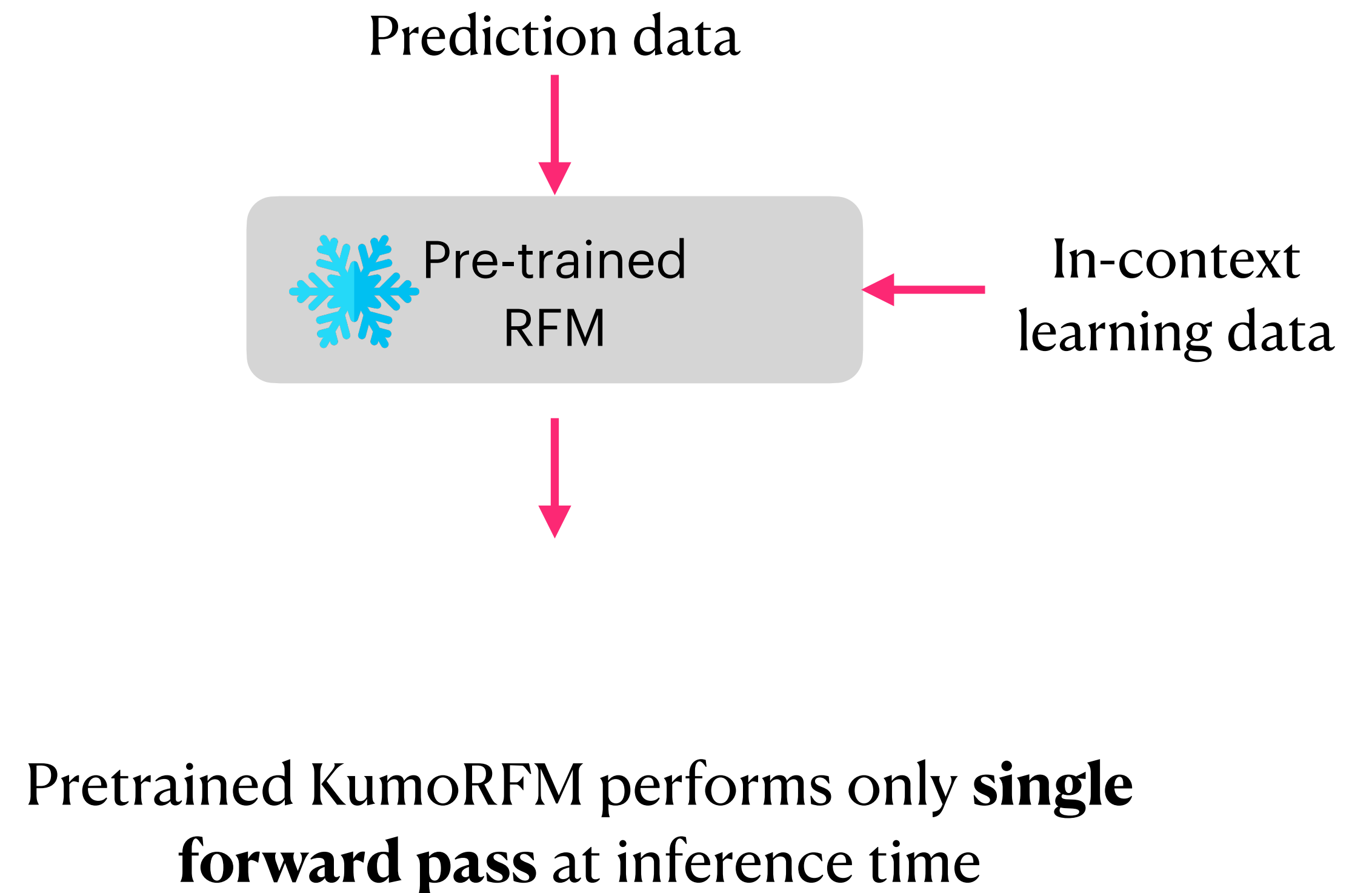
Predictions  
Embeddings

# Supervised vs. In-context learning

## Traditional modeling



## KumoRFM



# Capabilities of KumoRFM

- Supports ML tasks on relational data including classification, regression, and recommendation
- Adapts to unseen database schemas -> foundation model & architecture design
- Able to handle wide ranges of column types -> type aware encoders
- Offers insights into reasoning processes behind its prediction -> gradient-based saliency
- Efficiently runs predictions over billions of queries -> able to be fine-tuned

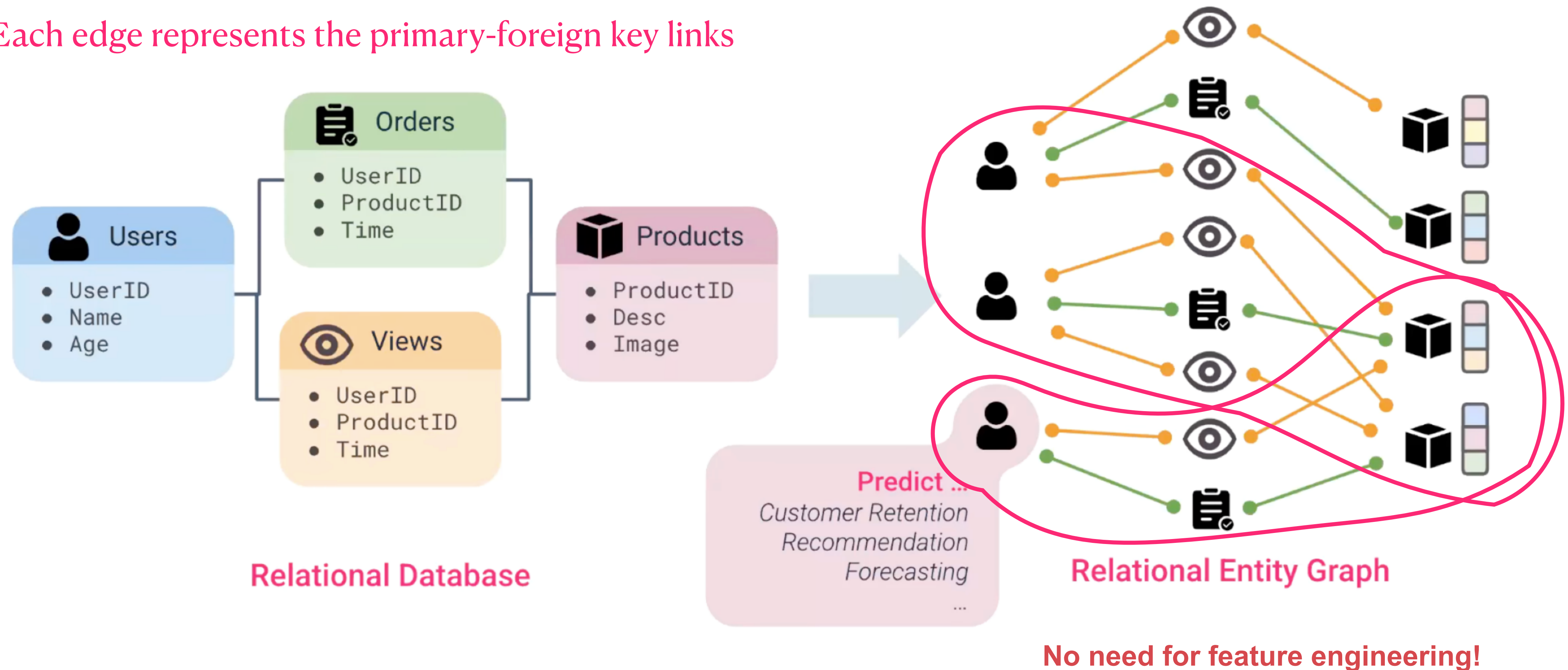
# Key Components of KumoRFM

- Relational Foundation Model
- Predictive Query Language
- In-context label generatator
- Explainability module
- Fine-tuning module

# How it works: Relational Deep Learning<sup>[1]</sup>

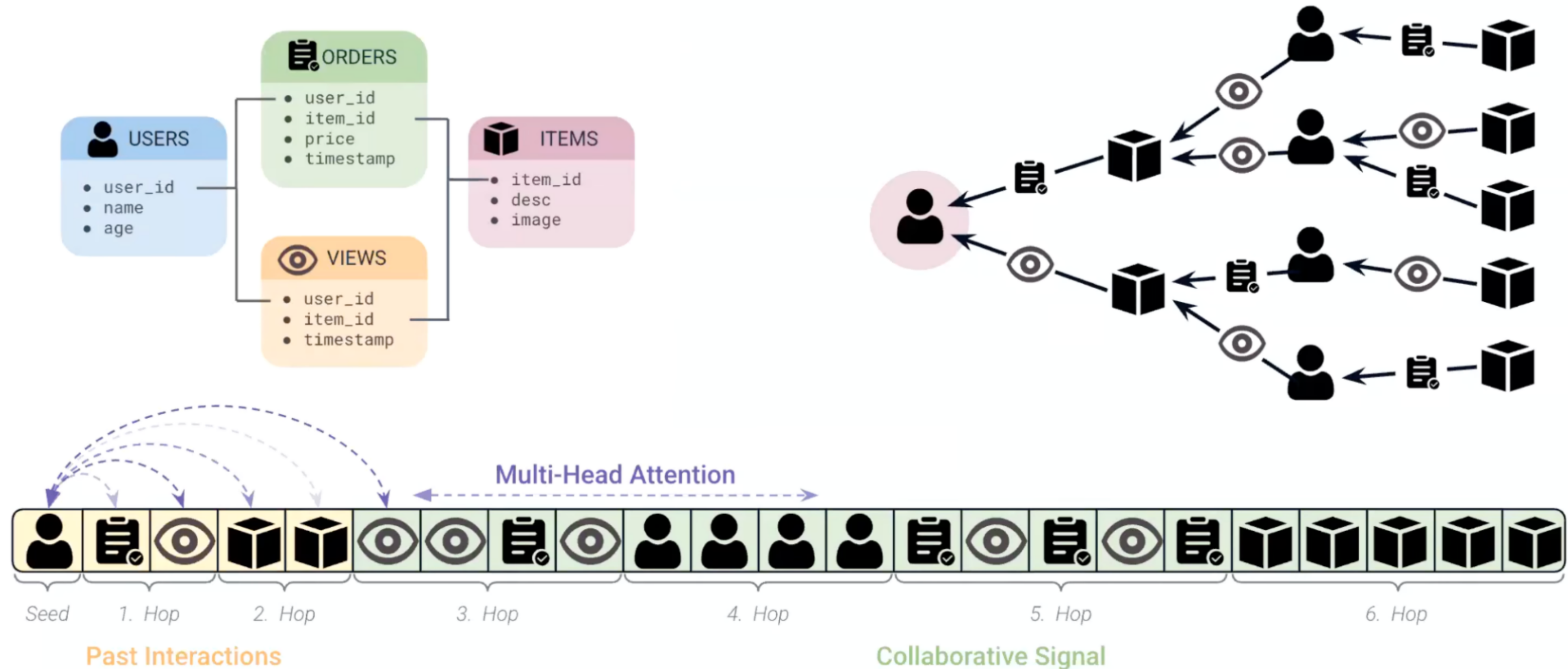
Each row is a node (entity)

Each edge represents the primary-foreign key links



# Architecture: Relational Graph Transformer<sup>[2]</sup>

**Tokenizes** the subgraph, attaches graph-specific **positional encodings**, and applies **multi-head attention**

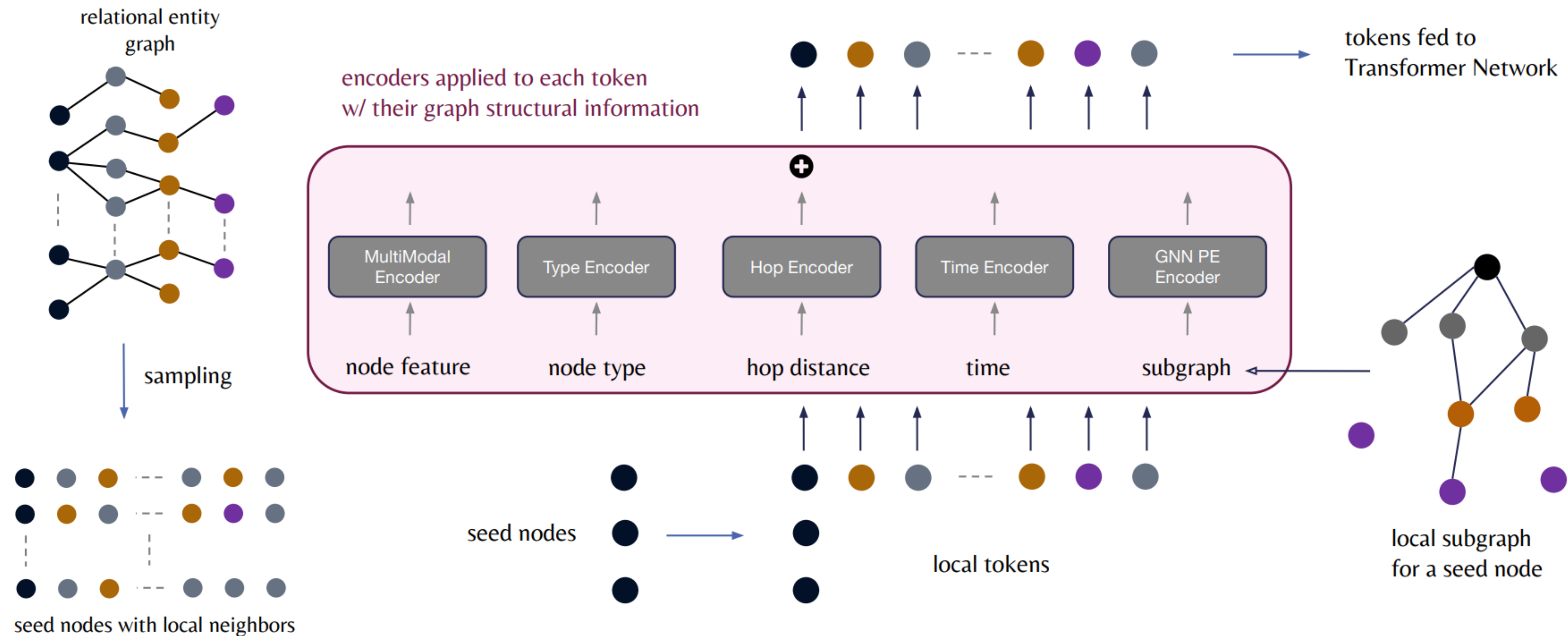


# Architecture: Relational Graph Transformer<sup>[2]</sup>

**KumoRFM has two model components**

1. Table-Invariant Row-level Encoder
  1. Encode each cell in every table  $T$  into a representation  $\mathbf{C}_T \in \mathbb{R}^{N_T \times C_T \times F}$
  2. Applies Transformer on the  $\mathbf{C}_T$  to obtain  $\mathbf{H}_T^{(L)} \in \mathbb{R}^{N_T \times F}$
  3. Handles diverse column types by semantic type, rather than just text

# Examples of Encoders



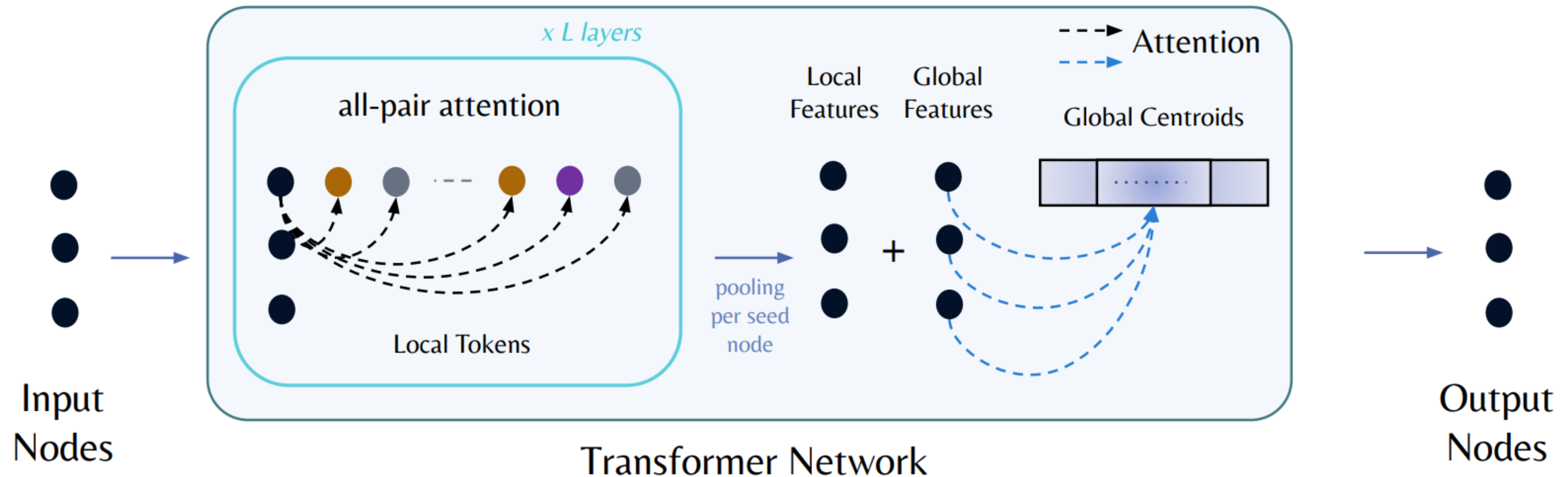
- **hop encoder:** captures the structural proximity between the entity node and other nodes in the subgraph.
- **time encoder:** encodes the relative time of facts/events w.r.t. the prediction time  $t$ .
- **subgraph encoder:** captures the local graph structure. This effectively captures parent-child relationships or important structural relationships such as cycles.
- ...

# Architecture: Relational Graph Transformer<sup>[2]</sup>

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  3. Handles diverse column types by semantic type, rather than just text
2. Relational Graph Transformer + In-context Learning
  1. One Transformer exchanges information within graphs
  2. One Transformer exchanges context and labels  $\left(\mathbf{H}_{\text{train}}^{(L)}, \mathbf{y}_{\text{train}}, \mathbf{h}_{\text{test}}^{(L)}\right)$  across graphs

# Transformer Network in Relational Graph Transfromer



**Local Module:** seed node  $v_i$  attend to all k-hop tokens via **full self-attention** (not just to adjacent ones like in GNNs)

$$h_{local}(v_i) = Pool(FFN(Attention(v_i, \{v_j\}_{j=1}^K)))_L$$

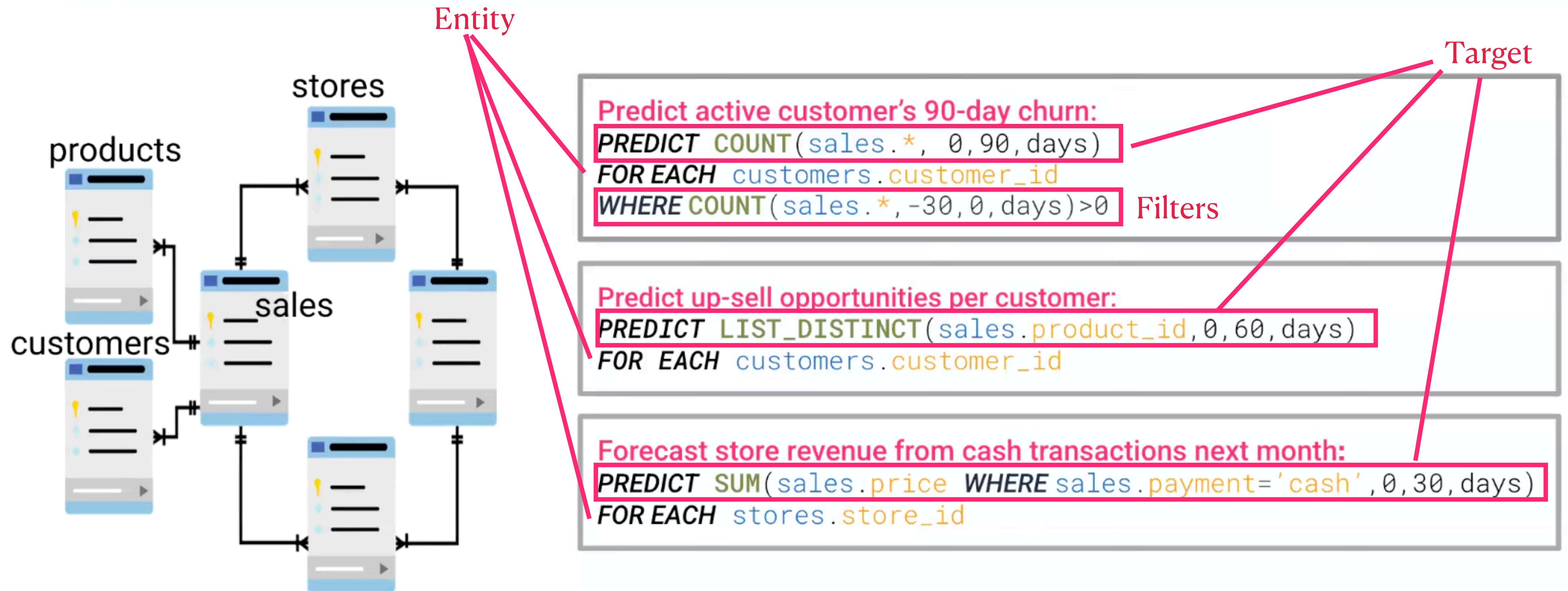
**Global Module:**  $v_i$  attends to B learnable **global centroid tokens** (updated via EMA K-means over the training mini-batches)

$$h_{global}(v_i) = Attention(v_i, \{c_b\}_{b=1}^B)$$

# Key Components of KumoRFM

- Relational Foundation Model
- **Predictive Query Language**
- In-context label generatator
- Explainability module
- Fine-tuning module

# Kumo Prompt: Predictive Query Language<sup>[3]</sup>



# Key Components of KumoRFM

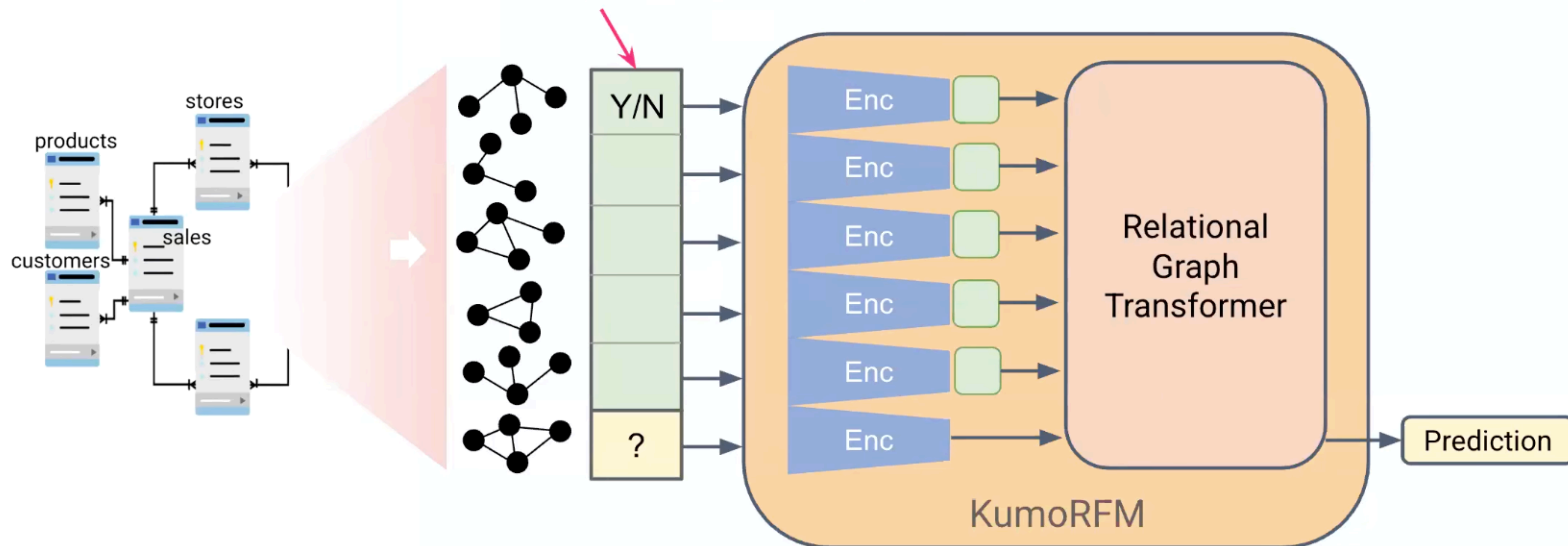
- Relational Foundation Model
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# KumoRFM: In-Context Learning

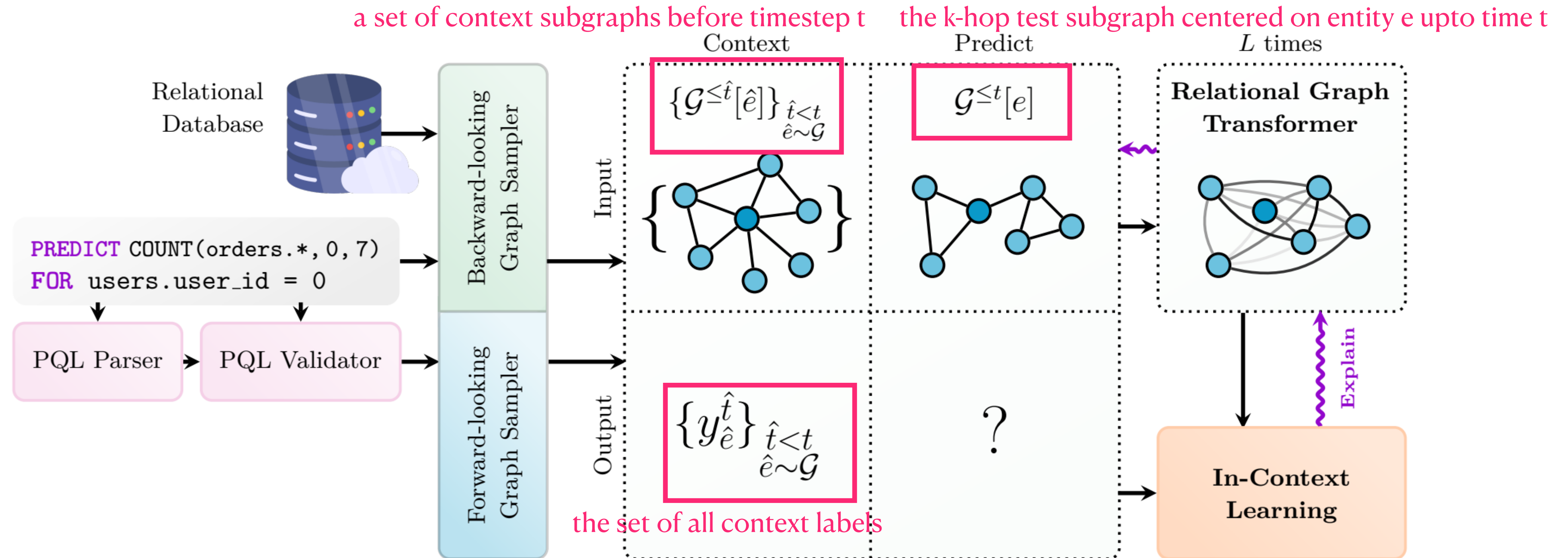
Predictive Query → "Prompt"

*PREDICT* **COUNT**(TRANSACTIONS, 0, 30, days)=0

*FOR EACH* CUSTOMERS.CUSTOMER\_ID



# Overview of KumoRFM



**Backward-looking graph sampler:** builds context and test subgraphs by looking at database up to a given timestamp

**Forward-looking graph sampler:** retrieve ground-truth labels

**Node Attribute prediction (classification/regression):** a property/label of an existing node

**Link prediction (recommendation):** whether an edge (relationship) will exist between two nodes

# Key Components of KumoRFM

- Relational Foundation Model
- Predictive Query Language
- In-context label generatator
- **Explainability module**
- Fine-tuning module

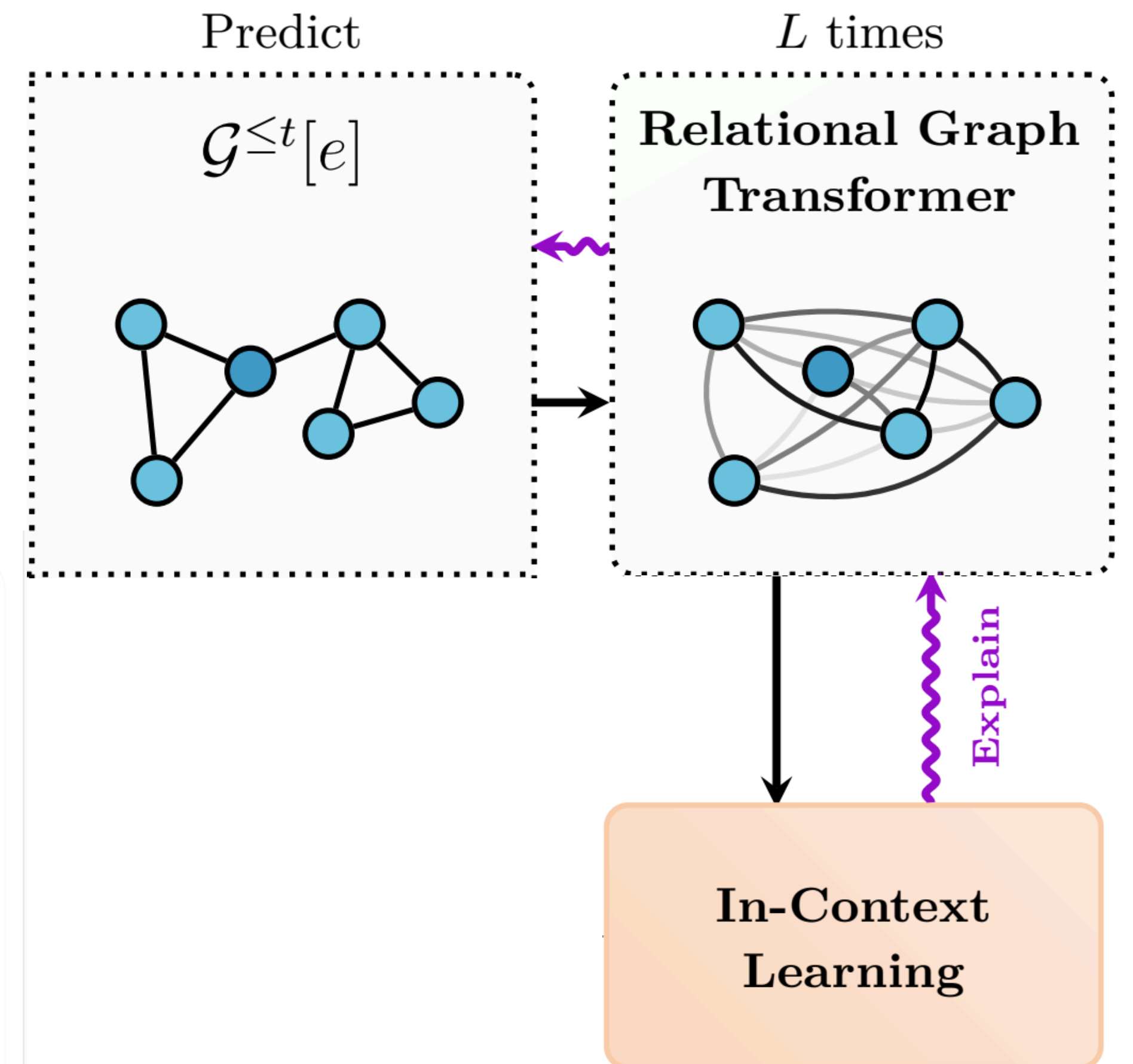
# KumoRFM Explainability

- KumoRFMs can be fully-differentiable and enable gradient-based explanation techniques
- It computes the gradient of the model's prediction score with respect to each input feature
- Importance scores are computed on the cell level rather than on the feature level
- Conversion to textual summary by an LLM

**PREDICT** COUNT(orders.\*, 0, 30) > 0 **FOR** users.user\_id=1

*The model predicts that the user has a moderate likelihood of placing at least one order in the upcoming month. Key factors influencing this prediction include:*

- **Order Count:** Users with only a few past orders have a very low likelihood of ordering soon, while those with more orders show increased probabilities.
- **Order Date Recency:** Recent orders (6-12 months ago) greatly increase the chance of placing new orders soon.
- **Fashion News Frequency and Club Membership:** Users who regularly receive fashion news or have active club membership status show higher probabilities of ordering.



# Key Components of KumoRFM

- Relational Foundation Model
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- **Fine-tuning module**

# Fine-tuning KumoRFM

Motivation: In-context learning is not scalable for business level predictions over billions of entities

Fine-tuning refers to the process of specializing KumoRFM to a single dataset and a single task.

## **Steps:**

- Replace its table-invariant encoders with dataset-specific ones
- Substitute the final in-context learning head with a task-specific one
- Follow the general RDL blueprint to fine-tune the model in a supervised fashion based on a pre-generated training table.

# Experiments

KumoRFM is trained on a mixture of  
**publicly available relational databases and synthetic data**  
For label generation, they utilize a random Predictive Query generator

KumoRFM is evaluated on



7 relational databases

30 predictive tasks (classification, regression recommendation)

| Dataset    | Domain     | #Tasks |
|------------|------------|--------|
| rel-amazon | E-commerce | 7      |
| rel-avito  | E-commerce | 4      |
| rel-event  | Social     | 3      |
| rel-f1     | Sports     | 3      |
| rel-hm     | E-commerce | 3      |
| rel-hm     | Social     | 5      |
| rel-trial  | Medical    | 5      |
| Total      |            | 30     |

KumoRFM has not seen any RelBench datasets during its pre-training phase, which guarantees no leakage!

Baselines

- **LightGBM:** Supervised ensemble of decision trees
- **Data Scientist:** An expert data scientist that solves each task by manual feature engineering
- **RDL:** End-to-end supervised GNN
- **LLM:** A Llama 3.2 3B model that is asked to do in-context predictions

[4] Robinson, J., Ranjan, R., Hu, W., Huang, K., Han, J., Dobles, A., ... & Leskovec, J. (2024). Relbench: A benchmark for deep learning on relational databases. Advances in Neural Information Processing Systems, 37, 21330-21341.

# Entity Classification

| Dataset    | Task            | SUPERVISED |                |              | FOUNDATIONAL |                                      |              |
|------------|-----------------|------------|----------------|--------------|--------------|--------------------------------------|--------------|
|            |                 | LightGBM   | Data Scientist | RDL          | LLM          | KumoRFM<br>(in-context) (fine-tuned) |              |
| rel-amazon | user-churn      | 52.22      | 67.60          | <b>70.42</b> | 62.55        | 67.29                                | <b>70.47</b> |
|            | item-churn      | 62.54      | 81.80          | <b>82.81</b> | 73.41        | 79.93                                | <b>82.83</b> |
| rel-avito  | user-visits     | 53.05      | —              | 66.20        | 53.36        | 64.85                                | <b>78.30</b> |
|            | user-clicks     | 53.60      | —              | 65.90        | 54.07        | 64.11                                | <b>66.83</b> |
| rel-event  | user-repeat     | 53.05      | —              | 76.89        | 53.36        | 76.08                                | <b>80.64</b> |
|            | user-ignore     | 79.93      | —              | 81.62        | 68.65        | <b>89.20</b>                         | <b>89.43</b> |
| rel-f1     | driver-dnf      | 68.86      | 69.80          | 72.62        | 80.03        | <b>82.41</b>                         | <b>82.63</b> |
|            | driver-top3     | 73.93      | 82.40          | 75.54        | 87.11        | <b>91.07</b>                         | <b>99.62</b> |
| rel-hm     | user-churn      | 55.21      | 69.00          | 69.88        | 63.81        | 67.71                                | <b>71.23</b> |
| rel-stack  | user-engagement | 63.39      | 90.30          | 90.59        | 81.23        | 87.09                                | <b>90.70</b> |
|            | user-badge      | 63.43      | 86.20          | 88.86        | 79.99        | 80.00                                | <b>89.86</b> |
| rel-trial  | study-outcome   | 70.09      | <b>72.00</b>   | 68.60        | 59.17        | 70.79                                | 71.16        |
| Average ↑  |                 | 62.44      | —              | 75.83        | 68.06        | <b>76.71</b>                         | <b>81.14</b> |

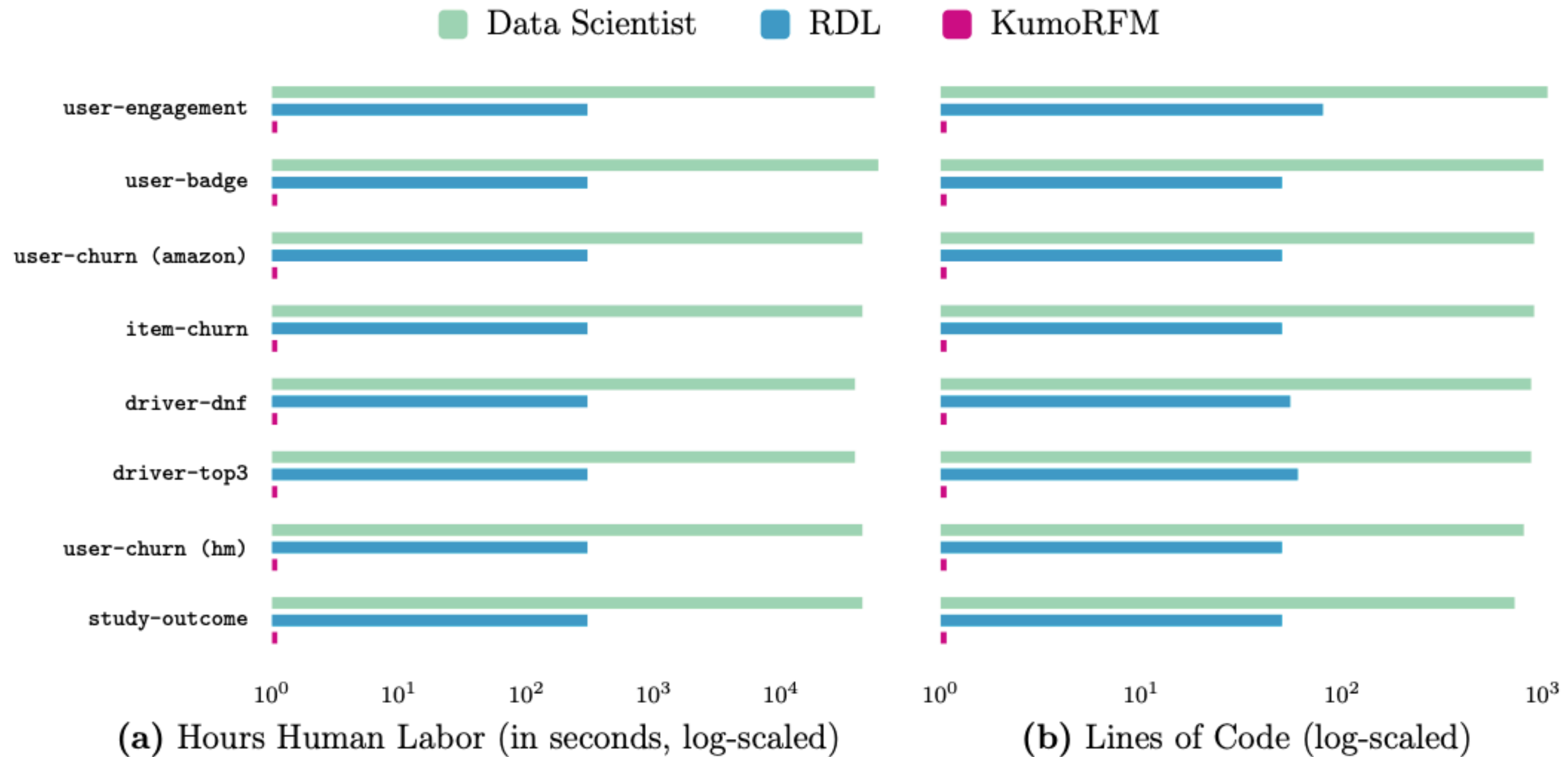
KumoRFM is on par with best baselines without finetuning, and outperforms them when finetuned

# Recommendation

| Dataset    | Task                  | LightGBM | SUPERVISED |        | FOUNDATIONAL |              |
|------------|-----------------------|----------|------------|--------|--------------|--------------|
|            |                       |          | RDL        |        | KumoRFM      |              |
|            |                       |          | GRAPHSAGE  | NFBNET | (in-context) | (fine-tuned) |
| rel-amazon | user-item-purchase    | 0.16     | 0.74       | 0.10   | <b>1.72</b>  | <b>2.93</b>  |
|            | user-item-rate        | 0.17     | 0.87       | 0.12   | <b>1.14</b>  | <b>2.25</b>  |
|            | user-item-review      | 0.09     | 0.47       | 0.09   | 0.22         | <b>1.63</b>  |
| rel-avito  | user-ad-visit         | 0.06     | 0.02       | 3.66   | <b>4.02</b>  | <b>4.17</b>  |
| rel-hm     | user-item-purchase    | 0.38     | 0.80       | 2.81   | 2.73         | <b>3.14</b>  |
| rel-stack  | user-post-comment     | 0.04     | 0.11       | 12.72  | 11.83        | <b>13.34</b> |
|            | post-post-related     | 2.00     | 0.07       | 10.83  | <b>11.80</b> | <b>12.21</b> |
| rel-trial  | condition-sponsor-run | 4.82     | 2.89       | 11.36  | 11.29        | <b>11.65</b> |
|            | site-sponsor-run      | 8.40     | 10.70      | 19.00  | <b>20.83</b> | <b>28.02</b> |
| Average ↑  |                       | 1.79     | 1.85       | 6.74   | <b>7.29</b>  | <b>8.82</b>  |

KumoRFM outperforms RDL without finetuning

# Time-to-First-Prediction



KumoRFM is 1,000 - 10,000 faster than alternatives

# Takeaways

What is it?    A foundation model for business data!

What can it do?    Make Zero-shot predictions on relational data!

Can we make it even better?    Fine-tune on a particular task!

Why does this matter?

- Predictions become commoditized
- Predictive models are democratized
- Prediction-driven applications become possible
- Data becomes even more important
- Fine-tuning empowers high performance use-cases